

Unit 4 — Optimization & Metaheuristics

Review deck for the test on Thu Oct 8

English for the exam · "ES:" tips in Spanish

Wiki: Optimization Basics · Gradient Descent · Simulated Annealing · Evolutionary Computation · GA · EA Representation & Selection · Swarm Intelligence · PSO · ACO · ABC

Optimization basics

- Find the input that **minimizes / maximizes** an objective (fitness) function. Only the **final state** matters (no path).
- Goal: the **global** optimum, without getting stuck in **local** optima.
- Every metaheuristic balances **exploration** (try new regions) and **exploitation** (refine good ones).
- Course function: **$f(\mathbf{x}, \mathbf{y}) = (\mathbf{x} + 2)^2 + (\mathbf{y} - 2)^2 + 10$** $\rightarrow$ minimum **$f = 10$ at $(-2, 2)$** (convex, one minimum).
- Benchmarks with many local minima: **Rastrigin, Ackley, Schaffer f6**.

The six algorithms at a glance

	GD	SA	GA	PSO	ACO	ABC
Who / year	Cauchy, 19th c.	Kirkpatrick et al. 1983	Holland 1975	Kennedy & Eberhart 1995	Dorigo et al. 1996	Karaboga 2007
Inspiration	calculus	annealing metals	evolution	bird flocks	ant pheromone	foraging bees
Solutions	1	1	population	swarm	colony	hive
Needs gradient	yes	no	no	no	no	no
Key operator	$w \leftarrow w - \eta \nabla f$	accept $e^{(-\Delta/T)}$	selection, crossover, mutation	$v \leftarrow w \cdot v + c_1 r_1 (p - x) + c_2 r_2 (g - x)$	$p \propto \tau^\alpha \eta^\beta$	$v = x + \phi(x - x_k)$
Typical problems	continuous, differentiable	both	both	continuous	combinatorial (TSP)	continuous
Global optimum guaranteed	only if convex	asymptotically (slow cooling)	no	no	no	no

Gradient descent

$$w \leftarrow w - \eta \cdot \nabla f(w)$$

Practice 1: $f = (x - 2)^2 + (y + 2)^2$, start $(0, 0)$, $\eta = 0.1$

- $\nabla f = (2(x-2), 2(y+2)) = (-4, 4) \rightarrow w_1 = \mathbf{(0.4, -0.4)}$, $f = 5.12$
- $w_2 = \mathbf{(0.72, -0.72)}$, $f = 3.2768 \cdot w_3 = \mathbf{(0.976, -0.976)}$, $f \approx 2.097$
- Each step multiplies the distance by $(1 - 2\eta) \rightarrow$ converges for $\mathbf{0 < \eta < 1}$; $\eta = 0.5$ in one step; $\eta = 1$ oscillates; $\eta > 1$ diverges.

ES: η pequeño = lento; η grande = se pasa o diverge. En funciones multimodales solo llega a un mínimo **local**.

Simulated annealing

- Pick a random neighbor; $\Delta = \text{cost}(\text{new}) - \text{cost}(\text{current})$.
- $\Delta < 0 \rightarrow$ **always accept**; otherwise accept with probability $e^{(-\Delta/T)}$.
- Cool down: $T \leftarrow \alpha \cdot T$.

$\Delta = 2$	$T = 10$	$T = 1$	$T = 0.1$
P(accept)	0.819	0.135	$\approx 2 \times 10^{-9}$

High $T \rightarrow$ random walk (exploration) · $T \rightarrow 0 \rightarrow$ hill climbing (exploitation).

ES: bug B2 del script de clase: $\alpha = 0.8$ con 100 000 iteraciones $\rightarrow T \approx 0$ en ~ 100 iteraciones. Usar $\alpha \approx 0.9999$.

The evolutionary loop

```
initialize population randomly
evaluate fitness of each individual
repeat until a stop condition:
    select parents          (fitter → more likely)
    recombine (crossover)   → offspring
    mutate offspring        (small random changes)
    evaluate offspring
    select survivors        → next generation
```

Dialects: **evolutionary programming** (L. Fogel, 1960s) · **evolution strategies** (Rechenberg & Schwefel, 1970s, real-valued) · **genetic algorithms** (Holland, 1975, bit strings).

GA: encoding and operators

- Assignment: 16 bits, sign-magnitude; `10000010000000010` → $x = -2$, $y = 2$, $f = 10$ (optimal).
- One-point crossover at 3: `110|10110` × `001|11001` → `11011001`, `00110110`.
- Bit-flip mutation: expected flips = $L \cdot p_m$ (typical $p_m \approx 1/L$).
- **Representation must match the operators:** permutations need swap / insert / scramble / inversion mutation and PMX / order / cycle / edge crossover — one-point crossover breaks permutations.
- Holland: **schemata** (e.g., `1**0*`) and **implicit parallelism** — short, fit building blocks multiply.

Selection

- **Fitness-proportional (roulette):** fitness [24, 23, 20, 11] → **0.31, 0.29, 0.26, 0.14**.
Problems: premature convergence early, lost pressure late, changes if you add a constant to f.
- **Linear ranking:** $P(i) = (2 - s)/\mu + 2i(s - 1)/(\mu(\mu - 1))$.
- **Tournament (size k):** pick k at random, keep the best — simple, no global fitness needed.
- **SUS:** one spin with λ equally spaced arms (less variance than roulette).
- **Survivors:** generational, steady-state, **elitism**, $(\mu + \lambda)$ parents + offspring, (μ, λ) offspring only.

PSO — particle swarm optimization

```
v ← w·v + c1·r1·(pbest - x) + c2·r2·(gbest - x)  
x ← x + v
```

inertia (momentum) + **cognitive** (own memory) + **social** (swarm's best)

Practice 7: $x = (1, 1)$, $v = (0.5, -0.5)$, $p = (0, 2)$, $g = (-2, 2)$, $w = 0.5$, $c_1 = c_2 = 1$, $r_1 = 0.5$, $r_2 = 0.2$

→ **$v = (-0.85, 0.45)$** , **$x = (0.15, 1.45)$** , $f: 20 \rightarrow \mathbf{14.925}$ (improved).

ES: el paper de 1995 no tiene w (coef. 2); la tarea usa la versión con inercia → di cuál usas (errata #7).

ACO — Ant System (Dorigo 1996)

- Probability of the next city: $p_{ij} \propto \tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}$, $\eta = 1/d$; tabu list = visited cities.
- Update after all ants: $\tau_{ij} \leftarrow \rho \cdot \tau_{ij} + \sum_k Q/L_k$ (ρ = persistence, $1 - \rho$ = evaporation).
- Paper's best: $\alpha = 1$, $\beta = 5$, $\rho = 0.5$, one ant per city.

Practice 8 ($\alpha = 1$, $\beta = 2$): weights B 0.25, C 0.125, D 1 $\rightarrow \mathbf{P = 0.18, 0.09, 0.73}$.

$\tau_{AB} = 0.5 \cdot 1 + 100/10 + 100/20 = \mathbf{15.5}$.

ES: *stigmergy* = comunicación indirecta a través del entorno (la feromona).

ABC and swarm intelligence

Artificial bee colony (Karaboga 2007):

- **Employed bees** exploit their food source (try a nearby position).
- **Onlookers** choose sources $\propto$ quality and exploit them.
- **Scouts** abandon a source after `limit` failed trials $\rightarrow$ random search (**exploration**).
- Parameters: number of sources (SN), limit, MCN.

Swarm intelligence: simple agents + local rules + information sharing $\rightarrow$ collective behavior. Millonas' 5 principles: proximity, quality, diverse response, stability, adaptability.

Which algorithm? (practice 9)

Problem	Choice	Why
Train neural-network weights	Gradient descent / backprop	differentiable, many dimensions
Shortest tour of 50 cities	ACO (or GA with permutations)	combinatorial graph problem
Minimize Rastrigin, no gradient	PSO , SA, real-valued EA	continuous, multimodal
Repair this week's timetable	Min-conflicts local search / CSP	few changes from a good start

ES: mutación, evaporación y abejas exploradoras cumplen el mismo papel: **mantener diversidad**.

Self-check (answer aloud, in English)

1. Write the PSO update and name each term.
2. What does SA do when T is very high? When $T \rightarrow 0$?
3. Why can't one-point crossover be used on permutations?
4. Why can roulette selection cause premature convergence?
5. Write the ACO transition rule; what do α and β control?
6. Which algorithms need a gradient?

Answers: [wiki study/practice-unit-4-optimization.md](#) · Algorithm Lab (Optimization tab) · Exam Drill quiz